

Phase 11 . 8 : Relational Image

The relational image $(R\Downarrow S)$ gives the image of set S under relation R.

Example 1 : Simple relational image

$$(R\Downarrow S)$$

Example 2 : Relational image with set literal

$$(R\Downarrow \{1, 2, 3\})$$

Example 3 : Relational image with singleton set

$$(parentOf\Downarrow \{john\})$$

Example 4 : Relational image with composition

$$((R \circ S)\Downarrow A)$$

Example 5 : Relational image with inverse

$$(R^\sim\Downarrow S)$$

Example 6 : Set comprehension with relational image (Solution 35 pattern)

$$[Person]$$

$$\mid \quad parentOf : Person \leftrightarrow Person$$

$$\{p : Person \mid p \in \text{dom } parentOf \bullet (p, (parentOf\Downarrow \{p\}))\}$$

Example 7 : Chained relational images

$$((R\Downarrow S)\Downarrow T)$$